

Canning fruits - demonstration lesson plan

A step-by-step guide for the volunteer leading the demo

Audience	Adults and teens (public outreach demonstration)
Group size	Any - works at a fair table or in a teaching kitchen
Time	About 45-60 minutes
Format	Live demonstration with narration and questions
You should be	An Experienced Master Food Preserver volunteer

New to leading a demo? Read this first.

You do not have to memorize anything. This guide is written to be followed in order - it is completely fine to hold it and read from it during the demo.

Your job is not to be a performer. It is to show the steps slowly, say why each one matters, and keep things safe. The bold black text tells you what to DO; the lines in quotes are examples of what to SAY.

Walk through the steps once on your own before the event so nothing surprises you. Lay your equipment out in the order you will use it.

Go slowly and narrate everything. Silence feels longer to you than to your audience. If you lose your place, just look back at the numbered step.

It is okay to not know an answer. Say "That is a great question - let me look it up in the OSU Extension MFP app so I give you the right answer," and move on. Never guess about safety.

Learning objectives

By the end of this demonstration, participants will be able to:

1. Explain why fruits can be safely canned in a boiling water bath (they are high-acid foods).
2. Use a tested chart to find the correct headspace, syrup option, and processing time for a fruit.
3. Describe how to handle jars, lids, and boiling water safely.
4. Find OSU/USDA tested recipes instead of relying on untested or family methods.

Agenda at a glance

The demo has five parts. Approximate timing - go at whatever pace your group needs:

- 1. Welcome & safety briefing (5 min)
- 2. Introduction: why water-bath canning works (5 min)
- 3. Live demonstration (12 steps) (25-30 min)
- 4. Questions & discussion (10 min)
- 5. Wrap-up & cleanup (5 min)

Before participants arrive - set up

do this first

Set up and inspect everything before anyone is watching. The full pack-and-inspect list is on the equipment checklist - bring these:

- Canning jars (pint or quart size, straight-sided recommended)
- Two-piece canning lids (flat lid and screw band)
- Large pot or boiling water canner with rack
- Jar lifter or tongs
- Ladle and funnel
- Wide spatula or plastic knife to remove air bubbles
- Clean towels or cloths
- Pot for heating syrup or water
- Timer or clock

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- Heat-resistant gloves or mitts

1. Welcome & safety briefing

read aloud, 5 min

Greet the group, introduce yourself as an OSU Master Food Preserver, and read these aloud word-for-word before you touch any hot equipment:

- Wash hands thoroughly before starting.
- Use clean, undamaged jars and follow manufacturer's instructions for lids.
- Avoid burns: handle hot jars and boiling water carefully with protective gloves or jar lifters.
- Always use tested recipes and processing times to prevent foodborne illness.
- Remove jars from hot water carefully to avoid splashing and burns.

2. Introduction - why water-bath canning works

5 min

Hold up a jar of home-canned fruit and ask the group what they think keeps it safe to store on a shelf. Let a few people answer, then explain it in your own words using these points:

Say something like:

"Fruits are high-acid foods. A boiling water bath - 212 degrees at sea level - gets hot enough to destroy the molds, yeasts, and enzymes that spoil fruit, and it pushes the air out so the jar seals."

"This only works for high-acid foods. Low-acid foods like vegetables, meats, and most mixtures are NOT safe in a water bath - they need a pressure canner. Today we are sticking to fruit."

"The whole process is safe because we follow a tested recipe exactly - the right headspace, the right time, adjusted for our altitude. I will point those out as we go."

3. Live demonstration

go slow, narrate the WHY, 25-30 min

Work through each step out loud. The bold line is what to do; "Why" is what to tell them; the red line is a safety point to say directly. Pause on each red line and make eye contact.

- 1. Prepare fruit: Wash fruit thoroughly. Peel, core, pit, slice, or halve as appropriate. To prevent darkening, soak cut fruit in ascorbic acid solution or anti-darkening product. Drain just before packing.**

Why it matters: Cleaning removes dirt and microbes; peeling and slicing prepare the fruit for packing. Ascorbic acid prevents enzymatic browning that affects color and quality.

Tell your audience: Handle knives carefully; avoid cross-contamination by washing hands and surfaces.

- 2. Prepare jars and lids: Wash jars in hot soapy water and rinse. Keep jars hot until ready to fill, either in hot water or in a warm oven. Prepare lids according to manufacturer's directions (usually simmering in hot water, not boiling).**

Why it matters: Hot jars reduce the risk of breakage and help maintain food temperature during filling and processing, which improves seal quality.

Tell your audience: Avoid sudden temperature changes to prevent jar breakage.

- 3. Choose packing method - hot pack or raw pack: Hot pack means heating fruit in boiling syrup, water or juice before packing; raw pack means packing raw fruit into jars and covering with boiling liquid.**

Why it matters: Hot packing helps expel air, shrinks fruit to prevent floating, and reduces processing time. Raw pack is quicker but may result in more floating fruit.

Tell your audience: Be cautious with hot liquids to prevent burns.

- 4. Prepare syrup or liquid: Prepare light, medium, or heavy syrup (as desired) by dissolving sugar in water and bringing it to a boil. Use syrup, water, or fruit juice to cover fruit in jars.**

Why it matters: Syrup preserves flavor and texture; boiling liquid helps kill microbes and pack fruit safely.

Tell your audience: Handle boiling liquids carefully.

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5. **Fill jars with fruit and cover with boiling syrup, water, or juice leaving appropriate headspace (usually 1/2 inch).**

Why it matters: Proper headspace allows for expansion during processing and vacuum seal formation.

Tell your audience: Use a funnel and handle jars carefully to avoid spills.

6. **Remove air bubbles by sliding a plastic spatula or blunt knife along the inside edges of the jar.**

Why it matters: Removing air bubbles prevents trapped air that can cause spoilage or poor seals.

Tell your audience: Avoid using metal knives to prevent jar damage.

7. **Wipe jar rims with a clean damp cloth to remove any residue.**

Why it matters: Ensures a clean sealing surface for lids to adhere properly.

Tell your audience: Use clean cloths to avoid contamination.

8. **Apply lids and screw on screw bands fingertip tight (not too tight).**

Why it matters: Allows air to escape during processing for proper vacuum sealing when cooling.

Tell your audience: Do not overtighten lids to avoid lid damage and poor seal.

9. **Process jars in a boiling water canner: Place jars on rack in the canner with at least 1 inch of water above jars. Bring water to a full rolling boil and process for the time specified by fruit type, jar size, and altitude (e.g., 20-30 minutes for apples or peaches).**

Why it matters: Boiling water kills harmful microorganisms and creates vacuum seals as jars cool.

Tell your audience: Use jar lifter to handle hot jars; avoid water splashing.

10. **After processing time, turn off heat and remove canner lid. Wait 5 minutes before removing jars from water.**

Why it matters: Prevents sudden temperature changes that could cause jar breakage or seal failure.

Tell your audience: Be cautious of steam when removing lid.

11. **Remove jars using jar lifter and place on towel or rack to cool undisturbed for 12-24 hours.**

Why it matters: Cooling allows vacuum seals to form and stabilizes the product.

Tell your audience: Do not retighten lids or disturb jars while cooling.

12. **Check seals: Press the center of each lid; it should be concave and not flex up and down. Remove screw bands and try to lift the lid with fingertips to confirm seal.**

Why it matters: Ensures that jars are properly sealed and product is safe for storage. Unsealed jars should be refrigerated and used promptly or reprocessed.

Tell your audience: Discard jars with bulging lids or signs of spoilage.

4. Questions & discussion

OSU-sourced answers, 10 min

Invite questions: "What questions do you have so far?" works better than "Any questions?" Below are the ones you will hear most, with answers you can trust. If you get a question not listed here, it is fine to look it up in the MFP app rather than guess.

Q: Can I use less sugar when canning fruit?

A: Fruits may be canned safely with reduced sugar or even with no sugar added using plain water. No adjustments are needed in the processing time, but absence of sugar may result in softer texture and less flavor. For jams and jellies, regular pectin products do not allow sugar reduction unless specified. Low- or no-sugar needed pectin varieties can be used for low-sugar or no-sugar jams or jellies. Reduced sugar in jams/jellies may result in a syrupy consistency; special pectins or alternative sweeteners should be used following manufacturer directions. Sugar is not necessary to prevent spoilage and can be omitted by replacing sugar syrup with water or unsweetened juice for canning fruits. Other sweeteners like honey or corn syrup can substitute some sugar but affect flavor and sweetness levels. Sucralose (Splenda) is heat stable and may be added before canning; other artificial sweeteners are best added after opening.

Source: SP 50 646 Preserving for special diets

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Q: Do I have to use a rack for canning?

A: Use a rack in the canner to keep jars off the bottom of the pan and prevent heat damage during processing. Use a canning rack or a substitute that allows water to circulate freely around the jars for even heat distribution and safe processing.

Source: PNW 199 Canning fruits

Q: How do I know a jar sealed?

A: After processing and cooling, jars that are sealed will have lids concave and will not flex up and down when pressed. Remove the ring band and test the seal by gently lifting the jar by the lid; properly sealed lids remain firmly attached. If the lid can be pressed or flexes, the jar is not sealed.

Source: SP 50 646 Preserving for special diets

Q: Can I reuse canning lids?

A: Canning lids are designed for one-time use only and should not be reused. Always use new, properly prepared lids for each canning session to ensure a safe seal.

Source: SP 50 646 Preserving for special diets

Q: What if my lid did not seal?

A: If jars do not seal after processing, the food may be refrigerated and consumed within a certain time, or re-processed using new lids within 24 hours following safe canning procedures. Alternatively, the unsealed jars can be frozen or refrigerated and used promptly. Discard jars if spoilage is evident.

Source: SP 50 646 Preserving for special diets

5. Wrap-up & cleanup

5 min

Close by reinforcing the main takeaways, then clean up:

- Recap what makes canned fruit safe: a tested recipe, the correct headspace, and the full processing time with altitude adjustment.
- Ask the group to name one thing they would do at home, or one thing that surprised them.
- Point them to help: "When in doubt, check the OSU Extension MFP app or call the Extension office - never guess a time or method."

Then clean up:

- Turn off heat and unplug equipment once cool.
- Wash and dry all equipment; store jars in a cool, dark place.
- Label sealed jars with contents and date; refrigerate any unsealed jars.

Sources

Every value above is drawn from these OSU/USDA Extension publications. Verify against the cited source before public use.

- EM 9446 Playing it safe when preserving elderberries
- PNW 199 Canning fruits
- SP 50 444 Preserving peaches
- SP 50 586 Preserving plums and prunes
- SP 50 646 Preserving for special diets
- SP 50 678 Preserving pears
- SP 50 757 Fruit pickles chutney
- SP 50 780 Preserving berries
- SP 50 883 Preserving cherries